MILITARY SPECIFICATION

CORD ASSEMBLY, DETONATING

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the detonating cord assembly critical item, hereinafter referred to as the assembly.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 2024.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1377

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MILITARY

MIL-B-85251	Bomb, Binary Chemical, BLU-80/B.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-453	Radiographic Inspection.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1168	Ammunition Lot Numbering.
MIL-STD-45662	Calibration System Requirements.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

X4900676	Cord Assembly, Detonating.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

American Society for Testing and Materials

ASTM-E498-73	Standard Methods of Testing for Leaks Using the Mass Spectrometer Leak Detector or Residual Gas Analyzer in the Tracer Probe Mode.
ASTM-E499-73	Standard Methods of Testing for Leaks Using the Mass Spectrometer Leak Detector in the Detector Probe Mode.
ASTM-F78-71	Standard Method for Calibration of Helium Leak Detectors by use of Secondary Standards.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS.

3.1 Item description. The assembly consists of four mild detonating cords affixed into an aluminum sleeve with a common initiating charge and cover. The opposite end of each cord has an output charge and cover.

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3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Initiation and output. When the assembly is initiated with a MARK 339 MOD 1 Fuze or an approved equivalent energy source, all four boosters shall detonate and the output of each booster shall be sufficient to produce a minimum dent 0.013 inch deep in a steel witness plate (see Figure 1).

3.2.1.2 Temperature shock. The assembly shall withstand alternating exposure to temperatures of -48 degrees Celsius (°C) for 90 minutes, +75°C for 30 minutes, and +254°C for 10 minutes without degradation of performance, when tested at +155°C, and shall not constitute a safety hazard after exposure to temperature shock.

3.2.1.3 Low temperature. The assembly shall withstand exposure to a temperature of -62°C for 14 days without degradation of performance.

3.2.1.4 High temperature (storage). The assembly shall withstand exposure to a temperature of +74°C for 14 days without degradation of performance.

3.2.1.5 High temperature (safety). The assembly shall withstand exposure to a temperature of +99°C for 2 hours followed by +280°C for 30 minutes without constituting a safety hazard and shall not detonate, deflagrate, or burn.

3.3 Design and construction.

3.3.1 Production drawings. The assembly shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents listed on Drawing X4900676 and as specified herein.

3.3.1.2 Mechanical integrity. Each assembly shall be free from voids, inclusions, or impurities in any of the explosive components and the internal construction of the assembly shall conform to Drawing X4900676.

3.3.1.3 Leak rate. Each assembly shall have a leak rate of less than 1×10^{-5} atmospheric cubic centimeters of helium per second at a differential pressure of 1 atmosphere and a temperature of +25 $\pm$5°C.

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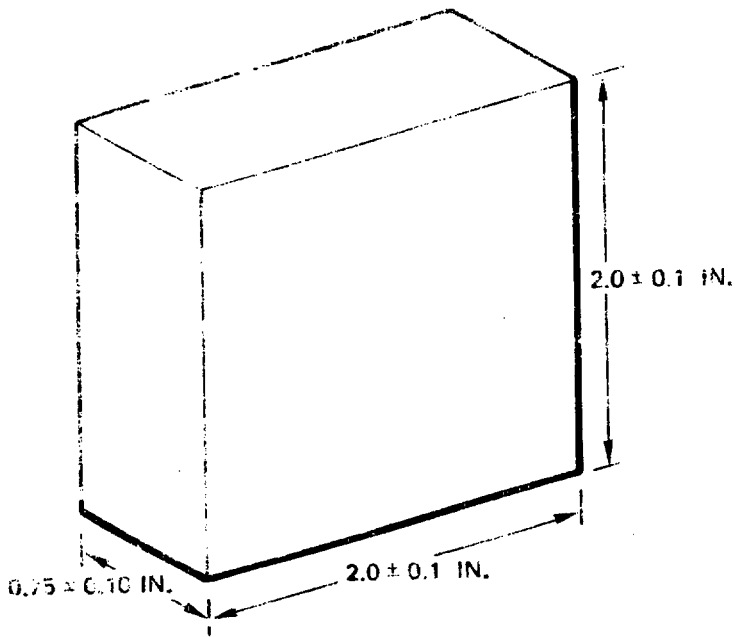


FIGURE 1. Witness plate.

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3.3.2 Standards of manufacture.

3.3.2.1 Certification. When specified in the contract or purchase order (see 6.2.2), the contractor shall include written certification, accompanied by objective quality evidence as defined in MIL-STD-109, that the materials, processes, and parts used in the assembly meet the applicable requirements for each item.

3.3.2.2 Working environment. Any special working environments for the manufacture of the assemblies shall be as specified in MIL-B-85251.

3.3.2.3 Manufacturing processes and procedures. When specified in the contract or purchase order (see 6.2.2), the contractor shall prepare, in the contractor's own format, written procedures covering the detailed equipment and process requirements and procedures which the contractor proposes to use and does use in the manufacture of assemblies to this specification. Copies of such work instructions, together with referenced documents, shall be made available to the procuring activity in the manner specified in the contract or purchase order. Examination and review of such documents will not constitute technical approval by the procuring activity, but will evidence only that the documents exist and are in sufficient detail to describe the process and procedures used in the manufacture of assemblies to this specification.

3.4 First article. Unless otherwise specified in the contract or purchase order, the contractor shall funish a preproduction sample of 15 assemblies to the testing activity designated in the contract or purchase order for first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor. After approval of the preproduction sample, changes in material, parts, procedures, design, or processes, may be cause for resubmission of a preproduction sample.

3.4.1 Retest. At the discretion of the procuring activity, preproduction tests, or any portion thereof shall be repeated under any of the following conditions:

a. The manufacturer has modified the product (such as a change of raw materials, processes, production procedures, or methods).

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b. Where there is evidence that the quality of the product has not been maintained. This evidence may be in the form of accumulated product failures, of systems failures attributable to the product, or failure of the product to pass any of the tests for production lot acceptance that may be conducted by or for the procuring activity.

c. When applicable documents have been amended or revised sufficiently so that continued validity of the previous preproduction testing is questionable.

d. When manufacturer has not produced assemblies to this specification for a period of 12 months or longer.

e. When the manufacturer changes the location at which cord assemblies are provided.

3.5 Documentation. When specified in the contract or purchase order (see 6.2.2), inspection reports shall be prepared as specified in 4.1.2.2 and 4.2.4.

3.6 Workmanship. The device shall be fabricated in a manner that will ensure compliance with all requirements of this specification and the assembly drawing. All parts shall be free from chips, dirt, grease, rust and other foreign materials. There shall be no burrs or other imperfections of manufacture which will adversely effect the assembly, safety, or functioning of the product. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used. Assembly shall proceed in a careful, neat, and orderly manner to prevent either damage to the parts, or the inclusion of foreign materials and defective or marginal parts and subassemblies.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspections (see 4.1.2.1).

b. Quality conformance inspections (see 4.2).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspections. Preproduction inspections shall be conducted on the preproduction sample (see 3.4.1) and shall consist of the examinations and tests of Table I in the order shown.

4.1.2.1.1 Acceptance criteria. The failure of one or more assemblies to pass any of the preproduction tests shall cause rejection of all preproduction samples.

TABLE I. Preproduction inspections.

Examination or test	Requirement paragraph	Method paragraph
Group I (10 units)		
Visual/radiographic/ leakage inspection	3.3.1, 3.3.1.2, 3.3.1.3, 3.6	4.4.1, 4.4.2, 4.4.7
Low temperature	3.2.1.3	4.4.4
High temperature (storage)	3.2.1.4	4.4.5
Temperature shock/initiation and output	3.2.1.2	4.4.3
Group II		
High temperature (safety) - 5 units	3.2.1.5	4.4.6

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4.1.2.2 Preproduction inspection report. When specified in the contract or purchase order (see 6.2.2), results of the preproduction inspections shall be prepared for the procuring activity. The report shall contain the following information accompanied by a certification (see 6.2.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the preproduction sample complies with all quality assurance provisions of this specification for the assembly.

b. Number of units of product inspected.

c. Results obtained for all inspections performed, including copies of data on dent depths.

d. Contract or purchase order number and date, together with an identification and date of changes.

e. Certificates of compliance of all material procured directly by the contractor.

f. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certifications unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Quality conformance inspections. Quality conformance inspections shall be performed on each inspection lot and shall consist of the examinations and tests listed in Table II in the sequence shown.

4.2.1 Inspection lot definition. Inspection lot definition shall be in accordance with MIL-STD-105. An inspection lot shall be formed from assemblies which have passed Group I inspections. An inspection lot shall not contain more than 200 assemblies. Lot numbering shall be in accordance with MIL-STD-1168.

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TABLE II. Quality conformance inspections.

Examination or test	Requirement paragraph	Method paragraph
Group I (100%)		
Visual/radiographic/ leakage inspection	3.3.1, 3.3.1.2, 3.3.1.3, 3.6	4.4.1, 4.4.2, 4.4.7
Group II (sample)		
Temperature shock/initiation and output	3.2.1.1, 3.2.1.2	4.4.3

4.2.2 Sample. A sample of 15 assemblies shall be randomly selected from each inspection lot for Group II quality conformance inspections.

4.2.3 Acceptance criteria. Failure of one or more assemblies in the sample to pass any of the Group II quality conformance inspections shall cause rejection of the inspection lot represented.

4.2.4 Quality conformance inspection report. When specified in the contract or purchase order (see 6.2.2), results of the quality conformance inspections shall be included with each lot. The report shall contain the following information accompanied by a certification which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of this specification for the assembly.

b. Number of units of product inspected.

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c. Results obtained for all inspections performed, including copies of data on dent depth.

d. Contract or purchase order number and date, together with an identification and date of changes.

e. Certificates of compliance of all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be sustained by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certifications unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.3 Test equipment and conditions. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the contract or purchase order, specification, and drawing requirements at the rate of delivery specified in the contract or purchase order. The calibration of measuring and test equipment shall be in accordance with MIL-STD-45662. Test equipment, including associated calibration programs and operation and maintenance procedures are subject to disapproval by the Government representative. When special test equipment and circuits are devised or commercially available equipment is employed, all test equipment circuits and methods are subject to disapproval by the Government representative. Unless otherwise specified herein, the magnitude of any error introduced by test equipment shall not exceed 10 percent of the tolerance of the requirement being measured.

4.3.2 Test conditions. Unless otherwise specified herein, all tests shall be conducted at the test area ambient temperature and relative humidity. Where special conditions of tests are specified, equipment employed shall adequately provide specified conditions. All monitoring

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and measurement devices shall conform to the requirements of 4.3.1. When special conditions of tests are not imposed, commercially available direct-measurement equipment may be employed provided the requirements of 4.3.1 are met.

4.4 Inspection methods.

4.4.1 Visual inspection. The assemblies shall be visually inspected to determine conformance with the drawings, parts lists and other documents listed on Drawing X4900676. Assemblies not conforming to 3.3.1 shall be rejected.

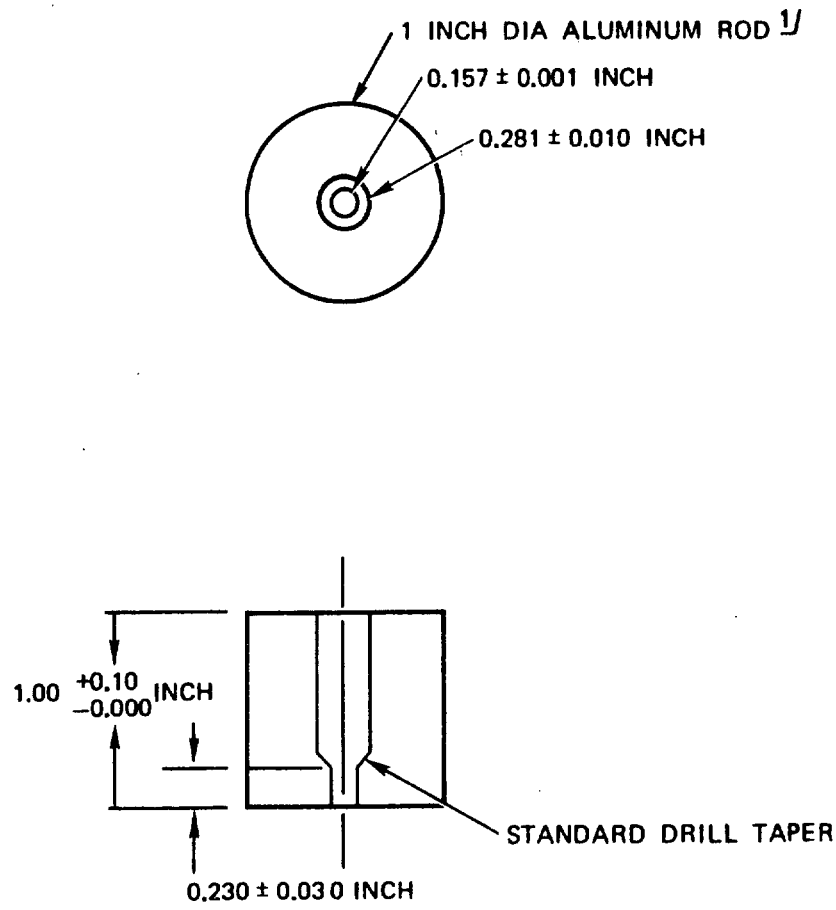
4.4.2 Radiographic inspection. Radiographic inspection of the assemblies shall be in accordance with MIL-STD-453. The assemblies shall meet the requirements of 3.3.1.2 to be acceptable.

4.4.3 Temperature shock/initiation and output. The assembly samples shall be placed in a temperature-conditioned chamber maintained at an internal temperature of -48 ±3°C for 90 minutes. At the end of the 90-minute period, the samples shall be transferred to a chamber temperature conditioned at +75 ±3°C. The transfer shall be made between the two chambers within 1 minute. The samples shall remain in the chamber for 30 minutes minimum. At the end of the 30-minute period, the samples shall be transferred to a chamber temperature conditioned at +254 ±3°C. The transfer shall be made between the two chambers within 1 minute. The samples shall remain in the chamber for 10 minutes minimum. The samples shall then be fired at +155 ±3°C. The assemblies shall be initiated in accordance with 3.2.1.1. The output of each booster shall be determined in accordance with MIL-STD-331, Test Method 301.1. The boosters shall be confined in an aluminum sleeve (see Figure 2). To be acceptable, the assembly samples shall meet the requirements of 3.2.1.1.

4.4.4 Low temperature test. The assembly samples shall be subjected to the low temperature test of MIL-STD-810, Method 502.1, except that the temperature shall be -62 ±3°C and the test time shall be no less than 336 hours.

4.4.5 High temperature (storage) test. Testing shall be conducted in accordance with MIL-STD-810. The assembly samples shall be placed in a temperature-conditoned chamber maintained at an internal temperature of +74 ±3°C for no less than 336 hours.

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1/MATERIAL: ALUMINUM ROD IN ACCORDANCE WITH QQ–A–225/6, T4 CONDITION

FIGURE 2. Confinement sleeve configuration.

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4.4.6 High temperature (safety) test. Testing shall be conducted in accordance with MIL-STD-810. The assembly samples shall be placed in a temperature-conditioned chamber maintained at an internal temperature of +99 ±3°C for 2 hours. At the end of the 2 hours, the samples shall be transferred to a chamber which is temperature conditioned to +280 ±3°C. The transfer between the two chambers shall be completed within 1 minute. The samples shall remain in the +280°C chamber for 30 minutes. To be acceptable, the assembly samples shall meet the requirements of 3.2.1.5.

4.4.7 Leak testing. All leakage determinations shall be made by a certified operator. The leak detector shall be calibrated in accordance with ASTM-F78 by a helium leak standard before and after testing a given lot of assemblies, or every 4 hours, whichever occurs first. Leak rate testing shall be conducted in accordance with ASTM-E498, Method A or ASTM-E499, Method A. The assemblies shall meet the requirements of 3.3.1.3.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C in accordance with FED-STD-102. The assemblies shall be preserved and packaged in a manner to assure delivery to the first destination without corrosion, deterioration, or damage to the cord. Packaging shall conform to the Code of Federal Regulations, 49 CFR 171-178, for shipment of Class A explosives.

5.2 Packing. Packing shall be Level C in accordance with FED-STD-102. The assemblies shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations for Level C. Packing shall conform to the Code of Federal Regulations, 49 CFR 171-178, for shipment of Class A explosives.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2.1), each package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The assembly is intended to be used to transmit a detonation signal from the fuze to the dissemination ports of the BLU-80/B binary chemical bomb.

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6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 15 assemblies as specified in 3.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, tests and approval of the first article.

c. Inspection requirements and test facilities (see 4.1).

d. Delivery schedule of parts and products (see 4.3.1).

e. Special markings (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited as follows:

Paragraph	Data requirement	Applicable DID
3.3.2.1, 4.1.2.2	Certification	DI-E-2121
3.3.2.3	Manufacturing instructions	DI-P-1614
3.5, 4.1.2.2, 4.2.4	Inspection reports	DI-T-2072

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.3 Safety precautions. The loading, assembly, and handling of the explosives, subassemblies, and finished items covered by this specification involve hazardous operations and require explosive safety precautions as stated in DoD 4145.26M.

Preparing activity:
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